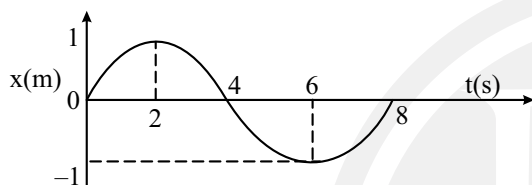


CHAPTER 13

Oscillations

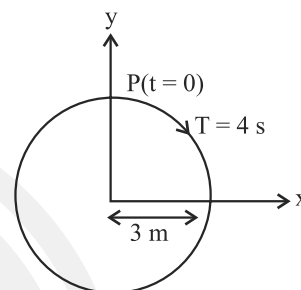
Periodic Motion, Superposition of Waves, Displacement, Velocity, Phase, Acceleration in SHM

1. The x - t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t = 2$ s is: (2023)



- a. $-\frac{\pi^2}{16} \text{ ms}^{-2}$ b. $\frac{\pi^2}{8} \text{ ms}^{-2}$
c. $-\frac{\pi^2}{8} \text{ ms}^{-2}$ d. $\frac{\pi^2}{16} \text{ ms}^{-2}$
2. Identify the function which represents a non periodic motion. (2022 Re)
- a. $\sin(\omega t + \pi/4)$ b. $e^{-\omega t}$
c. $\sin \omega t$ d. $\sin \omega t + \cos \omega t$
3. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is: (2020)
- a. $\frac{3\pi}{2}$ rad b. $\frac{\pi}{2}$ rad c. Zero d. π rad
4. Identify the function which represents a periodic motion. (2020-Covid)
- a. $\log_e(\omega t)$ b. $\sin \omega t + \cos \omega t$
c. $e^{-\omega t}$ d. $e^{\omega t}$
5. The displacement of a particle executing simple harmonic motion is given by $y = A_0 + A \sin \omega t + B \cos \omega t$. Then the amplitude of its oscillation is given by: (2019)
- a. $A_0 + \sqrt{A^2 + B^2}$ b. $\sqrt{A^2 + B^2}$
c. $\sqrt{A_0^2 + (A + B)^2}$ d. $A + B$
6. Average velocity of a particle executing SHM in one complete vibration is: (2019)
- a. $\frac{A\omega}{2}$ b. $A\omega$ c. $\frac{A\omega^2}{2}$ d. Zero

7. The radius of circle, the period of revolution, initial position and sense of revolution are indicated in the



y - projection of the radius vector of rotating particle P is: (2019)

- a. $y(t) = -3 \cos 2\pi t$, where y in m
b. $y(t) = 4 \sin\left(\frac{\pi t}{2}\right)$, where y in m
c. $y(t) = 3 \cos\left(\frac{3\pi t}{2}\right)$, where y in m
d. $y(t) = 3 \cos\left(\frac{\pi t}{2}\right)$, where y in m
8. A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is: (2017-Delhi)
- a. $\frac{\sqrt{5}}{2\pi}$ b. $\frac{4\pi}{\sqrt{5}}$
c. $\frac{2\pi}{\sqrt{3}}$ d. $\frac{\sqrt{5}}{\pi}$
9. When two displacements represented by $y_1 = a \sin(\omega t)$ and $y_2 = b \cos(\omega t)$ are superimposed, the motion is: (2015)
- a. Simple harmonic with amplitude $\frac{a}{b}$
b. Simple harmonic with amplitude $\sqrt{a^2 + b^2}$
c. Simple harmonic with amplitude $\frac{(a+b)}{2}$
d. Not a simple harmonic

10. A particle is executing S.H.M. along a straight line. Its velocities at distances x_1 and x_2 from the mean position are v_1 and v_2 respectively. Its time period is: (2015)

a. $2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$ b. $2\pi\sqrt{\frac{v_1^2 - v_2^2}{x_1^2 + x_2^2}}$
 c. $2\pi\sqrt{\frac{v_1^2 - v_2^2}{x_1^2 - x_2^2}}$ d. $2\pi\sqrt{\frac{x_1^2 + x_2^2}{v_1^2 + v_2^2}}$

11. A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β . Then, its time period of vibration will be: (2015 Re)

a. $\frac{2\pi\beta}{\alpha}$ b. $\frac{\beta^2}{\alpha^2}$
 c. $\frac{\alpha}{\beta}$ d. $\frac{\beta^2}{\alpha}$

12. The oscillation of a body on a smooth horizontal surface is represented by the equation,

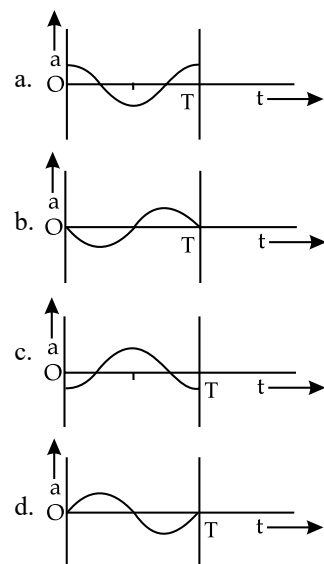
$$X = A \cos(\omega t)$$

Where,

X = displacement at time t

ω = frequency of oscillation

Which one of the following graphs shows correctly the variation a with t ? (2014)



Here a = acceleration at time t

T = time period

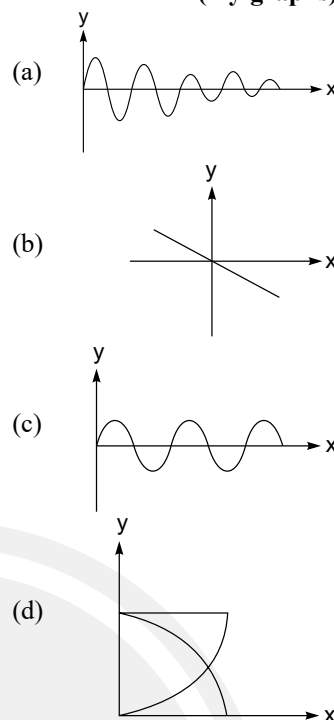
Energy in SHM (P.E., K.E. and T.E.)

13. A particle executing simple harmonic motion with amplitude A has the same potential and kinetic energies at the displacement (2024 Re)

a. $2\sqrt{A}$ b. $\frac{A}{2}$
 c. $\frac{A}{\sqrt{2}}$ d. $A\sqrt{2}$

14. Match List - I with List - II:

List-I (x-y graphs)



List-II (Situations)

- (i) Total mechanical energy is conserved
 (ii) Bob of a pendulum is oscillating under negligible air friction
 (iii) Restoring force of a spring
 (iv) Bob of a pendulum is oscillating along with air friction

Choose the correct answer from the options given below:

- a. (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
 b. (a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)
 c. (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
 d. (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

15. A body is executing simple harmonic motion with frequency ' n ', the frequency of its potential energy is: (2021)

- a. $2n$ b. $3n$ c. $4n$ d. n

Simple Pendulum and Loaded Springs

16. If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original

length, then the new time period of oscillation is $\frac{x}{2}$ time its original time period. Then the value of x is: (2024)

- a. $2\sqrt{3}$ b. 4 c. $\sqrt{3}$ d. $\sqrt{2}$

17. If the plate of a parallel plate capacitor connected to a battery are moved close to each other, then (2024)

- A. the charge stored in it. increases.
 B. the energy stored in it. decreases.
 C. its capacitance increases.
 D. the ratio of charge to its potential remains the same.
 E. the product of charge and voltage increases.

Choose the most appropriate answer from the given below:

- a. B, D and E only b. A, B and C only
c. A, B and E only d. A, C and E only

18. A simple pendulum oscillating in air has a period of $\sqrt{3}$ s. If it is completely immersed in non-viscous liquid, having

density $\left(\frac{1}{4}\right)^{\text{th}}$ of the material of the bob, the new period will be: (2023-Manipur)

- a. $2\sqrt{3}$ s b. $\frac{2}{\sqrt{3}}$ s c. 2 s d. $\frac{\sqrt{3}}{2}$ s

19. Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their means position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the means position is : (2022)

- a. 8 b. 11 c. 9 d. 10

20. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is: (2021)

- a. 6.28 s b. 3.14 s c. 0.628 s d. 0.0628 s

21. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is: (2018)

- a. 2s b. π s c. 2π s d. 1 s

22. A spring of force constant k is cut into lengths of ratio 1 : 2 : 3. They are connected in series and the new force constant is K' . Then they are connected in parallel and force constant is K'' . Then $K' : K''$ is: (2017-Delhi)

- a. 1 : 9 b. 1 : 11 c. 1 : 14 d. 1 : 6

23. A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3 s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m in kg is: (2016 - II)

- a. $\frac{16}{9}$ b. $\frac{9}{16}$ c. $\frac{3}{4}$ d. $\frac{4}{3}$

Answer Key

1. (a) 2. (b) 3. (d) 4. (b) 5. (b) 6. (d) 7. (d) 8. (b) 9. (b) 10. (a)
11. (a) 12. (c) 13. (c) 14. (c) 15. (a) 16. (d) 17. (d) 18. (c) 19. (b) 20. (c)
21. (b) 22. (b) 23. (b)

Mechanical Waves, Progressive and Stationary Waves; Intensity and Speed of Sound

- The displacement of a travelling wave $y = C \sin \frac{2\pi}{\lambda} (at - x)$ where t is time, x is distance and λ is the wavelength, all in S.I. units. Then the frequency of the wave is (2024 Re)
 - $\frac{2\pi\lambda}{a}$
 - $\frac{2\pi a}{\lambda}$
 - $\frac{\lambda}{a}$
 - $\frac{a}{\lambda}$
- A wave travelling in the +ve x -direction having displacement along y -direction as 1 m, wavelength 2π m and frequency of $1/\pi$ Hz is represented by: (2013)
 - $y = \sin(2\pi x + 2\pi t)$
 - $y = \sin(x - 2t)$
 - $y = \sin(2\pi x - 2\pi t)$
 - $y = \sin(10\pi x - 20\pi t)$

Superposition of Waves, Interference and Reflection of Wave

- A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2 . The ratio λ_2/λ_1 is: (2016 - I)
 - $\sqrt{\frac{m_1}{m_2}}$
 - $\sqrt{\frac{m_1 + m_2}{m_2}}$
 - $\sqrt{\frac{m_2}{m_1}}$
 - $\sqrt{\frac{m_1 + m_2}{m_1}}$

Vibration of String and Organ Pipe

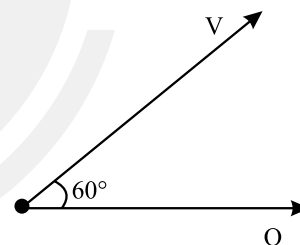
- The ratio of frequencies of fundamental harmonic produced by an open pipe to that of a closed pipe having the same length is (2023)
 - 3 : 1
 - 1 : 2
 - 2 : 1
 - 1 : 3
- The 4th overtone of a closed organ pipe is same as that of 3rd overtone open pipe. The ratio of the length of the closed pipe to the length of the open pipe is: (2023-Manipur)
 - 8 : 9
 - 9 : 7
 - 9 : 8
 - 7 : 9

- An organ pipe filled with a gas at 27°C resonates at 400 Hz in its fundamental mode. If it is filled with the same gas at 90°C, the resonance frequency at the same mode will be: (2022 Re)
 - 512 Hz
 - 420 Hz
 - 440 Hz
 - 484 Hz
- If the initial tension on a stretched string is **doubled**, then the ratio of the initial and final speed of a transverse wave along the string is : (2022)
 - 1 : 2
 - 1 : 1
 - $\sqrt{2} : 1$
 - 1 : $\sqrt{2}$
- The length of the string of a musical instrument is 90 cm and has a fundamental frequency of 120 Hz. Where should it be pressed to produce fundamental frequency of 180 Hz? (2020-Covid)
 - 60 cm
 - 45 cm
 - 80 cm
 - 75 cm
- The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the **open** organ pipe is: (2018)
 - 12.5 cm
 - 8 cm
 - 13.2 cm
 - 16 cm
- A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is: (2018)
 - 350 m/s
 - 339 m/s
 - 330 m/s
 - 350 m/s
- The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system? (2017-Delhi)
 - 20 Hz
 - 30 Hz
 - 40 Hz
 - 10 Hz
- The second overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe L meter long. The length of the open pipe will be: (2016 - II)
 - $\frac{L}{2}$
 - 4 L
 - L
 - 2 L
- An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is: (2016 - I)
 - 66.7 cm
 - 100 cm
 - 150 cm
 - 200 cm

14. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is: (2015)
a. 100 cm b. 120 cm c. 140 cm d. 80 cm
15. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The **lowest** resonant frequencies for this string is: (2015 Re)
a. 105 Hz b. 155 Hz c. 205 Hz d. 10.5 Hz
16. The number of possible natural oscillations of air column in a pipe closed at one end of length 85 cm whose frequencies lie below 1250 Hz are (velocity of sound = 340 ms⁻¹): (2014)
a. 4 b. 5 c. 7 d. 6
17. If n_1 , n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by: (2014)
a. $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$ b. $\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n_1}} + \frac{1}{\sqrt{n_2}} + \frac{1}{\sqrt{n_3}}$
c. $\sqrt{n} = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3}$ d. $n = n_1 + n_2 + n_3$
18. If we study the vibration of a pipe open at both ends, then the following statement is not true: (2013)
a. Pressure change will be maximum at both ends
b. Open end will be anti-node
c. Odd harmonics of the fundamental frequency will be generated
d. All harmonics of the fundamental frequency will be generated

Beats

19. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be: (2020)
a. 524 Hz b. 536 Hz c. 537 Hz d. 523 Hz
20. Three sound waves of equal amplitudes have frequencies $(n - 1)$, n , $(n + 1)$. They superimpose to give beats. The number of beats produced per second will be: (2016 - II)
a. 3 b. 2 c. 1 d. 4
21. A source of unknown frequency gives 4 beats/s, when sounded with a source of known frequency 250 Hz. The second harmonic of the source of unknown frequency gives five beats per second, when sounded with a source of frequency 513 Hz. The unknown frequency is: (2013)
a. 260 Hz b. 254 Hz c. 246 Hz d. 240 Hz
22. Two cars moving in opposite directions approach each other with speed of 22 m/s and 16.5 m/s respectively. The driver of the first car blows a horn having a frequency 400 Hz. The frequency heard by the driver of the second car is [velocity of sound 340 m/s]: (2017-Delhi)
a. 361 Hz b. 411 Hz c. 448 Hz d. 350 Hz
23. A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 ms⁻¹. Then, the frequency of sound that the observer hears in the echo reflected from the cliff is: (2016 - I)
(Take velocity of sound in air = 330 ms⁻¹)
a. 765 Hz b. 800 Hz c. 838 Hz d. 885 Hz
24. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 ms⁻¹ at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 ms⁻¹) is: (2015 Re)



- a. 106 Hz b. 97 Hz c. 103 Hz d. 100 Hz
25. A speeding motorcyclist sees traffic jam ahead him. He slows down to 36 km hour⁻¹. He finds that traffic has eased and a car moving ahead of him at 18 km hour⁻¹ is honking at a frequency of 1392 Hz. If the speed of sound is 343 ms⁻¹, the frequency of the honk as heard by him will be (2014)
a. 1332 Hz b. 1372 Hz c. 1412 Hz d. 1454 Hz

Answer Key

1. (d) 2. (b) 3. (b) 4. (c) 5. (c) 6. (c) 7. (d) 8. (a) 9. (c) 10. (b)
11. (a) 12. (d) 13. (c) 14. (b) 15. (a) 16. (d) 17. (a) 18. (a) 19. (a) 20. (b)
21. (b) 22. (c) 23. (c) 24. (c) 25. (c)